

Salayhin Md Sirajus

Staff Data Engineer — Data Platform Architecture & Governance

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SUMMARY

Staff Data Engineer with 15+ years of experience, from software-engineering roots to sole architect and primary technical decision-maker of a company-wide data platform. Architected and operate a production **Apache Iceberg lakehouse on GCP** (Scala/Spark, BigLake, dbt) serving analytics, ML, and BI. Drove a platform-wide cost program cutting monthly spend ~60% (¥1.2M → under ¥500K).

Own governance and security for the data platform — PII masking at ingest, GDPR user-data erasure across Iceberg, access control as code, SOC 2 support. Designed a custom MCP server enabling governed, AI-assisted analytics on BigQuery. Previously operated batch and near-real-time telemetry pipelines for 100K+ daily trips at Pathao.

SKILLS

- **Languages:** Python, Scala, SQL, Bash
- **GCP:** BigQuery, BigLake, Cloud Storage, Pub/Sub, Dataproc, Dataflow, Cloud Composer
- **AWS:** S3, S3 Tables (Iceberg), Athena, DynamoDB, EMR, Lambda
- **Lakehouse & Modeling:** Apache Iceberg (production), Parquet, Medallion Architecture, Dimensional Modeling
- **ELT & Processing:** dbt Core / dbt Cloud, Apache Airflow, Apache Spark
- **Governance & Security:** PII masking, GDPR erasure, IAM-as-code (Terraform), BigQuery audit logs, data contracts, SOC 2 support
- **FinOps & Observability:** GCP Billing Export, Asset Inventory, Cloud Monitoring, data pipeline observability, cost & data freshness alerting
- **DevOps:** Terraform, GitHub Actions, Docker, Kubernetes
- **AI Tooling:** MCP servers, Claude Code, AI-assisted engineering
- **BI:** Tableau, Redash

EXPERIENCE

Zeals Co., Ltd — Tokyo, Japan

May 2021 – Present

Staff Data Engineer (promoted Nov 2023) · Senior Big Data Engineer (May 2021 – Oct 2023; joined remotely, relocated to Tokyo 2022)

- Sole architect and primary technical decision-maker for the company-wide data platform powering analytics, ML, and BI — owning architecture reviews, tooling decisions, and platform standards across the data organization.
- Architected and operate a production **Apache Iceberg lakehouse** on GCS (BigLake Metastore) with dual-engine access via BigQuery and Spark on Dataproc — chosen for cost, multi-engine reads without duplication, and protobuf schema evolution.
- Engineered Scala/Spark pipelines ingesting ~100 GB/day (peaks ~1 TB) of protobuf event streams from Pub/Sub through landing → bronze → gold medallion layers, with automated Iceberg table maintenance (compaction, snapshot expiry).
- Drove a platform-wide cost program spanning BigQuery optimization and the Iceberg migration, lowering monthly spend ~60% (¥1.2M → under ¥500K).
- Optimized BigQuery workloads using system tables and billing exports — partitioning, clustering, and gold-layer star-schema modeling (dim_/fact_ tables) — reducing query runtime by up to 90% and scan volume by 80%.
- Led adoption of dbt Core/Cloud with CI/CD, testing, and documentation standards — cutting data incidents reaching stakeholders from 2–3/month to under one every two months, and pipeline maintenance from 1–2 days/week to a few hours.
- Own platform governance and security: column-level PII masking at ingest backed by BigQuery policies, GDPR right-to-erasure across Iceberg tables and snapshots, IAM-as-code via Terraform with PR review (dedicated BigQuery IAM module), and partnered with security engineers on SOC 2 compliance for the analytics layer.
- Shipped the FinOps & observability suite: daily cost collection across all GCP projects with billing-based actuals, expensive-query and cache-miss alerts, BigQuery freshness and quota alerts, Airflow pipeline summaries, and a central observability dashboard.
- Designed a custom **MCP server for governed, AI-assisted analytics** on BigQuery — schema discovery and plain-language querying of business metrics, restricted to gold tables via a security layer; championed team AI adoption

(Claude Code enablement, best practices, custom skills, AI-assisted code review).

- Defined data contracts, quality standards, and SLAs across teams; mentored junior engineers.

Pathao Ltd — Dhaka, Bangladesh

Jun 2018 – May 2021

Senior Data Engineer Level 2

- Designed and operated batch and near-real-time telemetry pipelines (trip lifecycle, GPS location streams, driver/rider actions) for 100K+ daily trips across ride-sharing, courier, and food delivery — spanning multiple cities across Bangladesh and Nepal.
- Implemented data pipelines integrating transaction data from 20+ sources plus streaming data at high volume (~50 GB/day) and velocity (peaks of 10K events/sec) on Spark, Dataflow, GCS, and Airflow.
- Created the metrics framework computing 100+ business KPIs (driver earnings, leaderboards, operational metrics) with under-5-minute end-to-end latency.
- Spearheaded the company's data lake and 10 TB BigQuery data warehouse on Google Cloud Storage, halving data retrieval times.
- Established the analytics ecosystem on Dataproc, Dataflow, BigQuery, and Data Studio, and delivered a scheduled reporting framework generating 10+ financial and operational reports.

Augmedix Bangladesh Ltd. — Dhaka, Bangladesh

Jun 2015 – Mar 2018

Senior Data Engineer

- Core contributor to the design and build of a SQL Server 2016 data warehouse on AWS (S3-backed storage) supporting company-wide business reporting.
- Developed AWS data pipelines (S3, Lambda) consolidating data from third-party APIs, Google Sheets, and MySQL (100+ attributes), and a reporting suite producing 20+ operational reports.
- Delivered backend services serving millions of warehouse records to internal applications.

Nascenia Ltd — Dhaka, Bangladesh

Jun 2013 – May 2015

Software Engineer

- Launched a B2B agricultural marketplace platform with Ruby on Rails and MySQL.
- Developed a social media analytics and post-scheduling tool with Rails and PostgreSQL.

Right Brain Solution Ltd — Dhaka, Bangladesh

Nov 2011 – May 2013

Software Engineer

- Co-authored a high-traffic automotive portal for the Toronto Star.
- Created APIs for social media mobile apps and a multilingual B2C ski-rental e-commerce platform.

Earlier roles: Assistant Programmer, Nanosoft (2011) — PHP/MySQL appointment scheduling · Software Developer, Sitebloom (2010–2011) — Codelgniter school management system.

LANGUAGES

English — full professional proficiency · Japanese — beginner, actively studying (Genki I)

EDUCATION

IICT, BUET — Post Graduate Diploma in ICT

2019

Jagannath University — B.Sc in Physics

2008

CERTIFICATIONS

- [Data Engineering with Google Cloud Specialization](#) — Google Cloud Training, Coursera (2023)
- [Data Engineering, Big Data, and Machine Learning on GCP Specialization](#) — Google Cloud Training, Coursera (2023)